

THORSBY AGCM, AB, Canada

WMO: 715040

Lat: 53.2172N

Lon: 113.8947W

Elev: 754

StdP: 92.59

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-30.7	-27.4	-34.3	0.2	-30.5	-31.0	0.2	-27.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	27.6	16.6	26.0	16.1	24.4	15.5	18.9	24.8	17.7	23.6	16.7	22.6	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.8	13.1	22.0	15.4	12.0	20.7	14.3	11.1	19.4	57.0	24.7	52.9	23.6	49.7	22.7	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	-35.6	31.3	3.9	2.2	-38.4	32.9	-40.7	34.2	-42.9	35.4	-45.7	37.0
			WB	-35.7	20.9	3.9	1.7	-38.5	22.2	-40.8	23.2	-42.9	24.2	-45.7	25.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-9.0	-10.1	-4.4	3.1	10.5	14.2	16.5	15.6	11.0	4.1	-4.0	-10.7		
	DBStd	11.66	8.63	8.61	7.82	5.51	4.39	2.92	2.66	3.04	4.49	4.66	7.22	8.01		
	HDD10.0	3153	588	563	447	212	48	3	0	1	41	188	420	642		
	HDD18.3	5573	846	797	706	456	245	127	69	94	223	440	670	901		
	CDD10.0	649	0	0	0	6	63	129	201	174	69	7	1	0		
	CDD18.3	26	0	0	0	0	1	3	12	9	1	0	0	0		
	CDH23.3	582	0	0	0	2	56	68	200	169	85	2	0	0		
	CDH26.7	96	0	0	0	0	6	7	34	32	18	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	486	20	15	14	22	52	84	97	75	48	23	17	18		
	PrecMax	594	55	40	39	57	161	196	237	190	145	62	49	41		
	PrecMin	379	1	1	2	1	8	15	34	6	4	2	3	5		
	PrecStd	59	16	9	10	16	31	44	49	44	28	16	10	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	8.1	12.6	22.8	27.5	27.5	29.9	29.7	29.0	21.9	15.2	7.0		
		MCWB	3.0	3.4	6.4	10.8	12.8	16.6	17.5	17.6	16.0	12.0	7.6	2.7		
	2%	DB	6.3	5.1	9.2	18.6	25.0	25.3	27.3	27.2	25.7	17.3	10.2	4.6		
		MCWB	2.3	1.4	4.0	8.7	11.9	15.3	17.9	17.1	14.5	9.4	5.0	1.1		
	5%	DB	4.6	3.3	6.6	14.9	22.4	23.5	25.7	25.2	22.9	14.6	6.7	2.6		
		MCWB	1.4	0.4	2.6	6.4	11.2	14.4	17.4	17.0	13.7	7.8	2.7	-0.4		
	10%	DB	2.9	1.6	4.4	12.1	19.9	21.6	23.7	23.2	20.0	12.0	4.7	0.5		
		MCWB	0.1	-0.7	1.1	5.4	10.1	13.5	16.5	16.2	12.4	6.4	1.6	-1.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	3.7	6.8	11.0	14.1	18.4	21.0	20.3	16.7	12.1	8.0	3.1		
		MCDB	7.3	7.7	12.0	22.3	24.1	25.3	26.9	25.3	25.4	20.7	14.5	6.6		
	2%	WB	2.5	1.7	4.4	8.8	12.9	16.5	19.2	18.8	15.3	10.0	5.4	1.3		
		MCDB	5.8	4.6	8.6	17.9	22.0	22.7	24.9	24.7	24.3	15.9	9.4	4.2		
	5%	WB	1.5	0.6	2.8	7.2	11.9	15.3	18.1	17.6	14.0	8.3	3.4	-0.2		
		MCDB	4.3	3.0	6.4	14.1	20.8	21.2	23.9	23.5	21.9	13.5	6.4	2.4		
	10%	WB	0.0	-0.7	1.3	5.6	10.9	14.3	17.1	16.5	12.8	6.9	1.8	-1.9		
		MCDB	2.4	1.7	4.1	11.2	18.8	20.1	22.6	22.5	19.1	10.9	4.3	0.4		
(35) Mean Daily Temperature Range	MDBR		9.8	10.7	10.3	11.2	13.2	11.7	12.2	12.7	13.2	11.1	9.4	9.4		
	5% DB	MCDBR	9.9	11.3	12.1	16.9	17.3	15.2	15.6	16.3	19.2	15.9	11.8	10.3		
		MCWBR	7.5	8.4	7.8	8.6	7.7	7.2	8.0	8.2	9.3	9.1	7.7	7.9		
	5% WB	MCDBR	10.1	11.1	11.7	16.0	15.2	13.3	14.1	14.5	17.6	14.8	11.2	10.7		
		MCWBR	7.6	8.4	7.7	8.5	7.2	7.0	7.8	8.1	9.0	8.6	7.5	8.6		
(40) Clear-Sky Solar Irradiance	taub		0.254	0.263	0.304	0.328	0.342	0.349	0.353	0.360	0.308	0.283	0.254	0.240		
	taud		2.321	2.379	2.363	2.384	2.390	2.387	2.421	2.356	2.494	2.547	2.464	2.319		
	Ebn at Noon		745	863	888	907	907	900	890	861	877	835	757	694		
	Edh at Noon		57	74	94	105	111	112	107	108	83	63	49	47		
(44) All-Sky Solar Radiation	RadAvg		0.99	2.02	3.51	4.66	5.65	5.98	6.17	5.08	3.52	2.05	1.01	0.73		
	RadStd		0.09	0.16	0.25	0.43	0.40	0.40	0.33	0.35	0.37	0.23	0.10	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (69 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page